

Circuit Rescue | Build the game

Regular CIRC class, stage 1 of two or three 35-minute contacts. Run each stage across a FULL five-day rotation before those same classes return. Each team keeps its board; six circuit sets are reused between classes.

One team tray

Covered switched 2-AA holder, two AA cells, active buzzer rated for 3 V, three insulated alligator leads, blunt conductive tool with insulated grip, cardboard, continuous foil rim, tape and large card tokens. Teacher pre-cuts openings.

The tool and rim are the switch

Battery + -> buzzer +. Buzzer - -> foil rim. Tool -> battery -. Contact closes the loop THROUGH the buzzer; moving away opens it. Never connect battery terminals directly. The buzzer signals a touch; there is no shock feature.

Pilot before class

Check exact voltage/polarity, covered batteries, secure separated connections, safe tool grip and reasonable volume. Power OFF for changes. Stop and disconnect if a part warms. No coin cells, mains, sharp probes or student soldering. Clips touch bare foil, not tape adhesive.

Stage 1	Minutes
Watch the touch; predict	0-3
Trace open/closed path and safety	3-8
Plan one large opening and matching token	8-12
Build, connect with power off, teacher check	12-25
Three contacts sound; three separated positions stay silent	25-30
Off, return electronics, store board	30-33
Explain where the switch is	33-35

No working parts: a partner says buzz on contact. Label this a paper simulation, not an electrical test. Large openings, pre-cut parts and a picture diagram provide support.

Circuit Rescue | Test and improve

Stage 2: retrieve the incoming class board and reconnect the shared tray with power off. Keep each before/after pair with the same player, hand, token, start position, tool and time limit except the one feature deliberately changed.

Stage 2	Minutes
Retrieve and teacher-check circuit	0-4
Agree on the fair-test rule	4-8
Three baseline trials; count separate rim contacts	8-15
Off; change opening size OR grip; predict	15-22
Same player and rule, three comparison trials	22-29
Off, return electronics, store evidence	29-33
Keep or undo, with evidence	33-35

Our test record

Version / one change	Trial 1	Trial 2	Trial 3
Baseline contacts	_____	_____	_____
Changed contacts	_____	_____	_____

Our fixed rule: _____

We changed: _____

We would keep / undo because: _____

Lift away between contacts so they can be counted. Choose the time limit from the pilot; 20 seconds is an estimate. Rotate roles after the before/after pair. Practice can affect results, so explain the limits of your conclusion.

Access and missing parts

Use a larger token/opening, supported grip, quiet simulation or a nonmoving role. Judge circuit reasoning and design evidence, not speed or fine-motor accuracy. A simulation remains labeled as such.

Circuit Rescue | Another user

Optional stage 3 makes this a three-rotation unit. Use it for unfinished tests, feedback and access improvements. Feedback from another player is not a controlled before/after trial.

Stage 3	Minutes
Retrieve and teacher-check	0-4
Explain the game and off switch	4-8
Watch another team; record one difficulty	8-16
Off; improve one access difficulty	16-24
Show the circuit path, result and change	24-29
Off; return all shared electronics	29-33
Explain what feedback changed	33-35

Operate the lesson

Teacher side: Playbooks > Circuit Rescue: build a buzzer game. Choose stage 1, 2 or 3, then Start class. Project Student directions. Next Step is manual. For the next class, choose Start this lesson for a new class and confirm. Reopening the same choice resumes it.

Keep the quantities small

For a class of 27, use six teams of four or five. Reuse six electronic trays, plus one teacher demonstration and one spare. Eight kits serve all classes. Store the cardboard boards for each class separately. Return electronics every period; individual take-home circuits are a separate budget decision.

Optional extensions

Use teacher-selected, checked math cards if useful. An optional Arduino teacher demonstration can count contacts after its exact model and input/driver wiring are checked. Do not connect this battery circuit directly to GPIO. A Raspberry Pi is not required.

Prepare and physically test the exact components before class. Technical background: learn.adafruit.com/low-tech-operation-game and adafruit.com/product/1536. Original classroom adaptation.

Canva | Logo practice | 35 minutes

Optional personal-interest logo practice from a template. Choose it when the class needs this skill; follow the current Playbook for the class's next activity. Show: "I can change a template to communicate one interest and make it readable when small."

Time	Say / do
0-4	Devices out. Open resources.svsd.net and school Canva.
4-8	Show two logos small. Ask what makes each easy to recognize. Model changing one icon, one font and two colors.
8-22	Choose a logo template. Change its colors, icon and font. Add a slogan of six words or fewer.
22-25	Zoom out to stamp size. Ask a partner what they can read. Fix one thing.
25-29	Share with your CIRC teacher's school account. The teacher opens it to confirm receipt. Keep public sharing unchanged.
29-35	Name one change you made. Save, close, return, plug in and count devices.

Done today

A shared logo that communicates an interest, includes a short slogan and reads when small. Paper and pencil with the same criteria is the complete fallback. Nobody needs to share private information.

Optional design challenge

Start from a blank Canva design. Make an original logo; get kind, specific feedback; explain two choices; share and verify. Choose this challenge only when it fits the class plan.

NeuroArt | Finish check and preview

Use for unfinished artwork when needed. Check the saved work before choosing this short finish check; it is not an added full-class unit.

Five-minute finish check

1. Open the actual saved picture or find the paper artwork.
2. Show one rounded crossing and a deliberate color choice.
3. Fix one unfinished corner or accidental color spill, if needed.
4. Save in the existing school location; reopen it. Keep the original.
5. Explain one choice by speaking, pointing or writing.

What students can say

"I rounded this corner so ____."

"I used these colors because ____."

"The change I made was ____."

Optional logo preview

Show a large logo, then make it small. Ask: "What survives when this gets tiny?" Connect NeuroArt choices about lines and color to clear logo choices. Use the Canva practice on the previous page if it fits the class plan.

Teacher recording storyboard, if you want a preview clip

0-10 seconds: show the saved artwork and say the goal.

10-30 seconds: zoom in on one corner and round it.

30-45 seconds: show one color change.

45-60 seconds: save, reopen and show the finished check.

60-75 seconds: show the logo preview and the three criteria: interest, slogan, reads small.

For a teacher preview, demonstrate with your own sample artwork and follow school recording and sharing rules.